Exp 5:-

Code:

```
# Hashing Using Chaining and Linear Probing
SIZE = 10
# Separate Chaining
chain_table = [[] for _ in range(SIZE)]

def chaining_insert(key):
    index = key % SIZE
    chain_table[index].append(key)



# Linear Probing
linear_table = [None] * SIZE

def linear_insert(key):
    index = key % SIZE

    for i in range(SIZE):
        new_index = (index + i) % SIZE

        if linear_table[new_index] is None:
            linear_table[new_index] = key
            return

    print("Hash table is full")



keys = [12, 22, 32, 15, 25]
```

```
for key in keys:
    chaining_insert(key)
    linear_insert(key)

print("Separate Chaining:")
for i in range(SIZE):
    print(i, ":", chain_table[i])

print("\nLinear Probing:")
for i in range(SIZE):
    print(i, ":", linear_table[i])
```

output

